

Finite Support of Projected Plancherel Weak Limits

A partial answer to Question (I) in arXiv:1712.04772

Automated mathematics run `fa_banach_001`

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Status. Candidate substantial partial result, likely valid. The theorem rules out all infinite-support limits. It does not classify the finitely supported limits.

1 Source question

The source is R. Adamczak, *Random non-Abelian G -circulant matrices. Spectrum of random convolution operators on large finite groups*, arXiv:1712.04772 [1]. Section 7, Question (I), page 25 asks for a characterization of the weak limits of projected Plancherel measures and notes that the examples in the paper contain only finite mixtures.

measure. This ends the proof of Theorem 6.2

□

7. FURTHER QUESTIONS

As shown in the preceding sections, random convolution operators on large finite groups exhibit a behavior, which combines features from the theory of large random matrices with independent entries, with those of classical probability theory. In our opinion, this is one of the reasons making this model interesting and worth investigating. Clearly there are multiple other aspects of the theory of random matrices, which we have not touched in the present work and which may potentially lead to interesting phenomena. Also, even the basic results obtained above give rise to new questions, both from the probabilistic and representation theory points of view. Below we list some of them.

- (I) As already mentioned in Section 2 we do not know how to characterize the class of measures μ on $\mathbb{Z}_+$, which are weak limits of projected Plancherel measures. In fact the class of examples we presented contains only finite mixtures. It would be of interest to see whether one can obtain more complicated examples.
- (II) While the limiting singular-value distribution follows a universal behaviour (as shown in Theorem 1.5), the eigenvalue distribution has been analysed in Theorem 1.10 only for Gaussian matrices. It is natural to conjecture universality also in this case. For sequences of groups with representations of bounded degrees this can be done by following the ideas of Meckes (which correspond to the small-dimensional representations part of our proof). For general groups we encounter a similar problem of instability of the spectral measure as in the classical circular law, moreover the model of random convolutions involves less independence (it is defined in terms of a smaller number of independent variables) than Wigner-like matrices. On the one hand, due to additive structure (Fourier transforms are sums of randomly weighted unitary matrices) this resembles somehow the

Figure 1: The complete source passage containing Question (I), rendered from page 25 of the source paper.

For a finite group G , write $\text{Irr}(G)$ for its irreducible complex characters and put

$$\mu_G = \sum_{\chi \in \text{Irr}(G)} \frac{\chi(1)^2}{|G|} \delta_{\chi(1)}.$$

This is a probability measure on $\overline{\mathbb{N}} = \mathbb{N} \cup \{\infty\}$, where $\overline{\mathbb{N}}$ has its one-point compactification topology.

2 Result and proof intuition

Proof intuition

The projected Plancherel measure remembers the commuting probability through one continuous moment:

$$\text{cp}(G) = \frac{|\text{Irr}(G)|}{|G|} = \int_{\overline{\mathbb{N}}} d^{-2} d\mu_G(d), \quad \infty^{-2} := 0.$$

Consequently, a limiting measure with any mass at finite integers forces a uniform positive lower bound on commuting probability. P. M. Neumann's theorem then puts the groups, uniformly, within bounded index of a group that is finite-by-abelian with bounded finite kernel.

The remaining point is not that the character degrees themselves are bounded — extraspecial groups show this is false — but that their *number of distinct values* is bounded. Clifford theory reduces each degree to a constituent over the bounded finite kernel and a projective representation of an abelian quotient. For a fixed multiplier, all irreducible projective representations of a finite abelian group have the same degree. Thus only boundedly many degree values occur. Support cardinality at most D is closed under weak convergence, which finishes the argument.

Theorem 1. *Let (G_j) be finite groups with $|G_j| \rightarrow \infty$. If μ_{G_j} converges weakly on $\overline{\mathbb{N}}$ to a probability measure μ , then μ has finite support.*

More quantitatively, if

$$c := \int_{\overline{\mathbb{N}}} d^{-2} d\mu(d) > 0,$$

then $|\text{supp } \mu|$ is bounded by a constant depending only on c .

Thus the infinite-support examples suggested in Question (I) do not exist. The exact characterization of the possible finite-support measures remains open.

3 The finite-by-abelian degree lemma

For a finite group K , let $\text{cd}(K) = \{\chi(1) : \chi \in \text{Irr}(K)\}$.

Lemma 2. *For every pair of positive integers a, b there is $D(a, b)$ with the following property. Suppose G is finite and has normal subgroups $H \leq \Gamma \leq G$ such that*

$$[G : \Gamma] \leq a, \quad |H| \leq b, \quad \text{and} \quad \Gamma/H \text{ is abelian.}$$

Then $|\text{cd}(G)| \leq D(a, b)$.

Proof. First consider Γ . Given $\eta \in \text{Irr}(\Gamma)$, choose an irreducible constituent $\theta \in \text{Irr}(H)$ of $\eta|_H$, and let $I = I_\Gamma(\theta)$ be its inertia group. Clifford correspondence gives

$$\eta = \text{Ind}_I^\Gamma \psi, \quad \eta(1) = [\Gamma : I] \psi(1)$$

for some $\psi \in \text{Irr}(I)$ lying over θ .

The character θ is I -invariant. The character-triple form of Clifford theory identifies the characters of I lying over θ with irreducible projective representations of the abelian group I/H for one fixed multiplier, with all degrees multiplied by $\theta(1)$. For a finite abelian group A and a fixed multiplier α , every irreducible α -projective representation has the same degree. Indeed, if R is the radical of the commutator bicharacter of α , the twisted group algebra $\mathbb{C}^\alpha[A]$ is a direct sum of equal matrix blocks of size $\sqrt{|A|/|R|}$.

It follows that, for each Γ -orbit in $\text{Irr}(H)$, all irreducible characters of Γ lying over that orbit have the same degree. Hence

$$|\text{cd}(\Gamma)| \leq |\text{Irr}(H)| \leq |H| \leq b.$$

Now take $\chi \in \text{Irr}(G)$ and a constituent $\eta \in \text{Irr}(\Gamma)$ of $\chi|_\Gamma$. A second application of Clifford theory gives

$$\chi(1) = te\eta(1),$$

where t is the size of the G -orbit of η and e is the degree of an irreducible projective representation of a subgroup of G/Γ . In particular, $1 \leq t \leq a$ and $1 \leq e \leq a$. Thus each of the at most b values of $\eta(1)$ gives at most a^2 values of $\chi(1)$. One may therefore take $D(a, b) = a^2b$. $\square$

4 Proof of the main theorem

Proof of Theorem 1. Define $f : \bar{\mathbb{N}} \rightarrow \mathbb{R}$ by $f(d) = d^{-2}$ for $d \in \mathbb{N}$ and $f(\infty) = 0$. This is continuous. The character-degree sum formula and the equality between the number of irreducible characters and the number of conjugacy classes give

$$\int f d\mu_G = \sum_{\chi \in \text{Irr}(G)} \frac{\chi(1)^2}{|G|} \frac{1}{\chi(1)^2} = \frac{|\text{Irr}(G)|}{|G|} = \text{cp}(G).$$

Therefore weak convergence implies $\text{cp}(G_j) \rightarrow c$.

If $c = 0$, positivity of f at every finite integer forces $\mu = \delta_\infty$, so the conclusion holds. Suppose $c > 0$. For all large j , $\text{cp}(G_j) \geq c/2$. P. M. Neumann's theorem, in the quantitative form quoted in [2, Theorem 1.2], supplies constants $a = a(c)$ and $b = b(c)$ and normal subgroups

$$H_j \leq \Gamma_j \leq G_j$$

such that $[G_j : \Gamma_j] \leq a$, $|H_j| \leq b$, and Γ_j/H_j is abelian. Lemma 2 shows that every μ_{G_j} is supported on at most $D(a, b)$ points.

It remains only to pass this bound to the weak limit. If μ had $D(a, b) + 1$ distinct support points, choose pairwise disjoint open neighborhoods of those points. Each has positive μ -measure. By the Portmanteau theorem, all these neighborhoods have positive μ_{G_j} -measure for all sufficiently large j , contradicting the support bound. Hence μ has finite support. $\square$

5 A mixed finite–infinite example

The conclusion cannot be strengthened to a dichotomy between finite support inside $\mathbb{N}$ and δ_∞ . Fix a prime p and let E_m be an extraspecial p -group of order p^{1+2m} . It has p^{2m} linear characters and $p - 1$ irreducible characters of degree p^m . Therefore

$$\mu_{E_m} = \frac{1}{p}\delta_1 + \frac{p-1}{p}\delta_{p^m} \implies \frac{1}{p}\delta_1 + \frac{p-1}{p}\delta_\infty.$$

This also explains why bounded commuting probability does not bound the degrees themselves; the bounded quantity needed in the proof is the number of distinct degree values.

6 Scope, novelty, and verification

The theorem settles the infinite-support part of Question (I), but it does not determine which finitely supported measures occur. In particular, arithmetic and representation-theoretic restrictions on their atoms and weights remain to be classified.

A bounded search covered the run’s solution, attempt, proof-gap, and registry indexes; the local parsed arXiv corpus; and targeted primary-source searches for projected Plancherel limits, character-degree distributions, and commuting probability. The search found Neumann’s structure theorem and work on scalar commuting-probability limits, but no explicit statement of Theorem 1. Novelty is plausible, not certified.

The proof is non-computational. Human review should focus on Lemma 2, especially the standard character-triple reduction and the equal-degree decomposition of a twisted group algebra of a finite abelian group. The source crop and every final packet page were rendered and visually inspected.

References

- [1] R. Adamczak, *Random non-Abelian G -circulant matrices. Spectrum of random convolution operators on large finite groups*, Electron. J. Probab. **24** (2019), paper no. 42; arXiv:1712.04772.
- [2] M. Tointon, *Commuting probability of infinite groups*, J. Lond. Math. Soc. (2) **101** (2020), 1280–1297; arXiv:1707.05565. See Theorem 1.2 for the finite-group theorem of P. M. Neumann used here.
- [3] I. M. Isaacs, *Character Theory of Finite Groups*, Dover Publications, 1994.
- [4] S. Eberhard, *Commuting probabilities of finite groups*, Bull. Lond. Math. Soc. **47** (2015), 796–808; arXiv:1411.0848.